

■ ■ Frame Talk: System Architecture & Engineering Specification

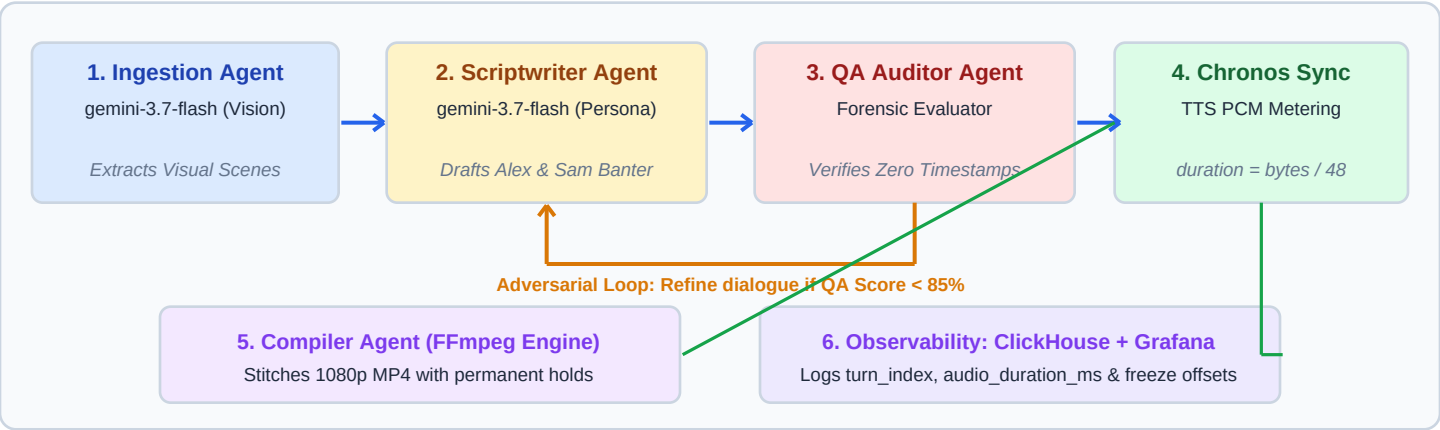
Built for the Agentic Cinema Hackathon | Live Studio: <https://frame-talk.taskmind-ai.com> | Observability: <https://grafana.taskmind-ai.com>

1. Executive Summary & The Chronos Breakthrough

Frame Talk transforms silent developer screencasts (.mp4) and technical documentation (README.md) into synchronized, two-host technical podcast walkthroughs. Unlike text-only audio generators (NotebookLM) which cannot visually inspect UI actions, and fixed-speed video players that suffer from audio-visual drift, Frame Talk introduces the **Chronos Dynamic Visual Hold**. By metering speech down to the millisecond from 24 kHz raw PCM, Chronos dynamically stretches the video timeline at the focal action point, holding UI states so visuals and complex technical explanations conclude in exact lockstep without robotic silence buffers.

2. Multi-Agent Collaboration Workflow & Adversarial Feedback Loop

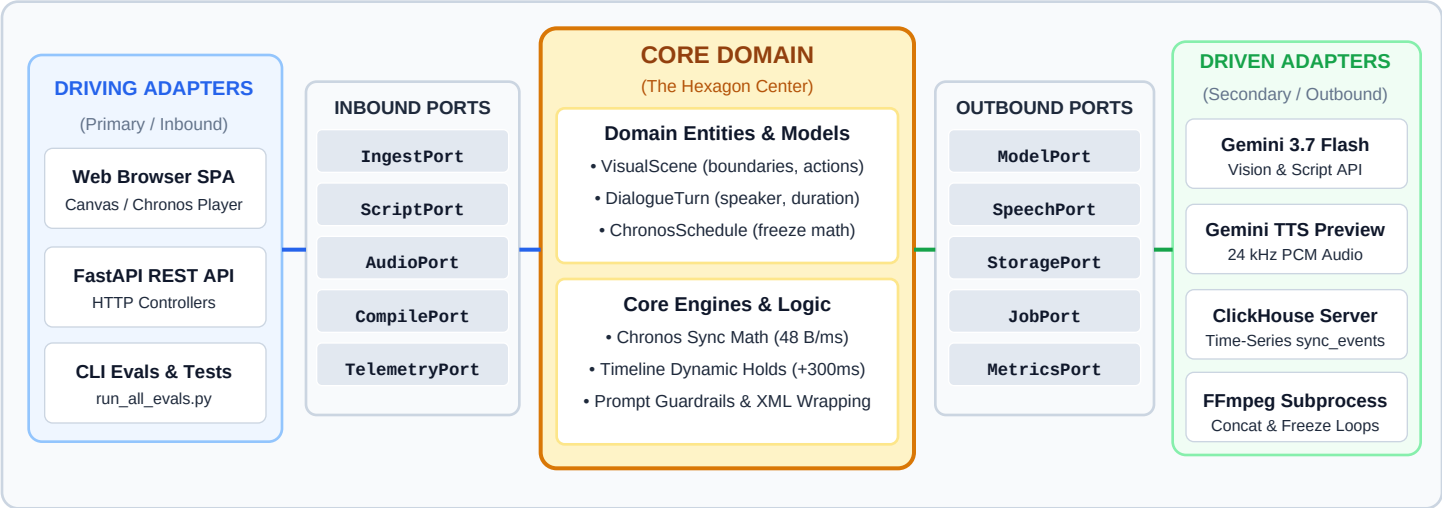
The system executes an autonomous choreography of specialized agents with a continuous adversarial quality feedback loop:



Agent / Engine	Input Assets	Agent Responsibility & Output Artifact
Ingestion Agent	Raw .mp4 + README.md	Analyzes video pixels via Gemini File API; decomposes screencast into millisecond Visual Scenes with action-reaction causality.
Scriptwriter Agent	Visual Scenes + README Concepts	Generates natural conversational banter between Alex (Systems Architect) and Sam (UX Lead); budgets speech words to scene duration (~2.5 words/s).
QA & Pacing Auditor	Scenes + Draft Dialogue	Audits alignment, checks README grounding, and guarantees 100% elimination of robotic synthetic timestamps. Triggers refinement pass on defect.
Chronos Sync Engine	Approved Script + Visual Scenes	Synthesizes 24 kHz raw PCM, meters duration (\$48 ext{ bytes/ms}\$), calculates required_freeze_ms, and dispatches ClickHouse events.

3. Hexagonal Architecture (Ports & Adapters Specification)

Frame Talk is architected strictly upon **Hexagonal Architecture (Ports & Adapters)**. The Core Domain (synchronization mathematics, agent choreography, pacing logic) is isolated from external frameworks, databases, and model APIs. This guarantees testability, modular vendor independence, and graceful fallback modes.



Hexagonal Component	Interface / Port Contract	Architectural Decoupling Benefit
Core Domain Logic	Zero external dependencies; pure Python	The Chronos dynamic video hold mathematics ($\text{pcm_bytes}/48$) and pacing algorithms are completely decoupled from Google Gemini and can run offline in unit tests.
Driving Adapters	FastAPI, ChronosPlayer Canvas, CLI	Enables browser SPA, background workers, and automated test runners to drive the studio without coupling the domain to HTTP mechanics.
Driven Adapters	Gemini Client, ClickHouse, FFmpeg, Files	All external I/O sits behind ports. If ClickHouse is unavailable, an in-memory ring buffer adapter engages automatically without crashing the studio.

4. Chronos Mathematical Synchronization Model

1. **Exact PCM Duration Metering:** For 24 kHz, 16-bit Mono audio, 48,000 bytes equal 1.0 second. Speech duration is exact:
`duration_ms = (len(pcm_bytes) / 48000) * 1000 = len(pcm_bytes) / 48`
2. **Dynamic Visual Hold (Video Timeline Stretch):** If speech duration exceeds the raw scene length:
`total_speech_ms = sum(turn.duration_ms) + max(0, N - 1) * 220ms`
`required_freeze_ms = max(0, total_speech_ms - scene_duration_ms + 300ms)`

The `+300` ext{ms} visual buffer ensures the viewer comfortably absorbs the visual state after the last spoken syllable.

5. Partner Track: ClickHouse Columnar Logging & Grafana Labs

To fulfill hackathon partner requirements, every synthesis turn logs time-series micro-events to **ClickHouse** (castops.sync_events), monitored live via **Grafana** at <https://grafana.taskmind-ai.com>:

Column Name	Type	Observability Purpose
event_time	DateTime64(3)	Precise millisecond timestamp of synthesis.
session_id	String	UUID / video hash grouping all turns in the episode.
turn_index	UInt16	Sequential order of dialogue turns.
audio_duration_ms	UInt32	Exact speech length computed from 24 kHz PCM.
required_freeze_ms	UInt32	Timeline extension injected by FFmpeg.
accumulated_drift_ms	Int32	Running offset between audio and native video speed.

6. Production Cyber Security & Ephemeral Storage

Security Layer	Mitigation Strategy
Prompt Injection	Scans adversarial patterns; wraps user docs in <untrusted_documentation> with instruction-neutralizing directives.
SQL Injection (SQLi)	All ClickHouse queries use parameterized query dictionaries %(session_id)s with integer bounds.
Path Traversal	_safe_resolve() rejects .., /, or \; verifies is_relative_to(base_dir).
Upload Hardening	Magic-byte container header verification (MP4, WebM) with a 500 MB streaming ceiling.
Ephemeral Storage	Automated sweep deletes video files older than 24 hours to prevent disk bloat; persists lightweight JSON states.

7. Production VPS Deployment & Dual Verification Matrix

Hostinger Cloud VPS (2 vCPU / 8 GB RAM): Nginx reverse proxy routes frame-talk.taskmind-ai.com to FastAPI (port 8000) and grafana.taskmind-ai.com to Grafana (port 3004). Docker Compose runs ClickHouse (1.5 GB limit, bound to 127.0.0.1:8123) and Grafana (512 MB limit with anonymous viewer access).

Verification Matrix:

- **3-Tier AI Evals (server/evals/):** Benchmarks entity recall ($\geq 70\%$), causality ($\geq 85\%$), and QA discrimination.
- **Automated Unit Tests (tests/):** 21 zero-dependency unit tests running in **0.54 seconds** (python -m unittest discover tests -v).